

Edgar Gabriel Caballero Torres

Data Scientist | Project Manager

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Summary

Data Scientist with experience in Machine Learning, predictive modeling, time series analysis, and end-to-end ETL pipelines in Google Cloud. Specialized in Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, and visualization (Power BI) to generate actionable insights that improve business decision-making. Known for clear communication with stakeholders, effective collaboration with cross-functional teams, and a strong focus on continuous improvement.

Skills

Programming & Queries: Python, SQL, JSON

Machine Learning & Deep Learning: Machine Learning, Scikit-learn, TensorFlow, Keras, PyTorch

Data Analysis & Visualization: Pandas | Polars, NumPy, Matplotlib | Seaborn, Jupyter Notebook, Power BI

Cloud & Infrastructure: ETL (Google Cloud Dataflow), APIs

Development & Tools: Git/GitHub

Languages: English (Professional), Spanish (Native)

Soft Skills

Clear and structured Communication (with technical and non-technical stakeholders) | Critical Thinking | Problem Solving | Innovation | Creativity | Time management & Self-management | Proactivity | Collaboration (in-person and remote) | Curiosity and Continuous Learning | Adaptability | Constructive Conflict Management | Communicate and collaborate with cross-functional teams | Entrepreneurial Mindset

Projects

Retail Customer Segmentation: BigQuery ML Pipeline [Local Repo | Cloud Repo]

Migration of a K-Means pipeline to a cloud-native architecture in BigQuery, optimizing retail customer segmentation for production scalability.

- Designed a Medallion Architecture (Bronze/Silver/Gold) in BigQuery to process ~540K transactions, automating ETL and SQL feature engineering.
- Implemented K-Means models directly in BigQuery ML, reducing data latency and performing "Segment Drift" analysis to track real-time consumer behavior shifts.
- Developed an interactive visualization suite (Plotly/Seaborn) featuring PCA to validate cluster stability and model separation.
- **Skills used:** BigQuery ML, Medallion Architecture, SQL (DML/DDI), Python (Pandas-GBQ), PCA, RFM Analysis.

Taxi Demand Forecasting | Time Series Analysis

Time series forecasting models to predict hourly airport taxi demand.

- Engineered temporal features (lags, rolling statistics) and performed time series decomposition to identify key daily/weekly patterns and peak demand hours.
- Developed and optimized a hybrid forecasting pipeline (ARIMA & ensemble models), achieving <10% RMSE for 1-hour-ahead predictions to inform driver allocation.
- **Skills used:** Data Engineering (ETL), Python Scripting, Unsupervised Learning (K-Means), RFM Analysis, Data Visualization (Plotly, Matplotlib).

A-B Test Evaluation | Conversion Impact

Designed and analyzed an A/B experiment to measure the impact of a new variant on user click-through conversions.

- Defined control (con) and experimental (exp) groups at the user level and validated random assignment and sample balance before analysis.
- Computed and compared click-through rates between groups, quantifying the absolute and relative uplift in conversions achieved by the experimental variant.
- **Skills used:** A/B Testing, Experimental Design, Hypothesis Testing, Conversion Rate Analysis, Statistical Inference, Data Cleaning & Aggregation.

Experience

PyAC S.A. de C.V., Project Manager | Construction Site Supervisor Mexico, June 2023 – June 2025

- Led high-investment projects (up to **35M MXN**), managing progress, cost, and quality metrics with an analytical approach to ensure compliance with specifications.
- Designed and optimized digital workflows (Revit–AutoCAD, Revit–BIMx), reducing rework and errors by **20%**, based on previous managing.
- Coordinated multidisciplinary teams, prioritizing tasks and solving problems under pressure, key skills for delivering end-to-end data solutions.

Ibero Steel, Project Manager Mexico, Dec 2022 – June 2023

- Led the end-to-end design of more than **4** standard padel courts.
- Produced over **10** marketing renders using V-Ray and SketchUp, improving client approval rates by **30%**, compared to previous Google Analytics data.
- Developed assembly manuals for prefabricated components, reducing on-site and manufacturing errors by **70%** based on first manufacturing batch.

Education

Cert. Tripleten, Data Science Mexico, Mar 2025 – Nov 2025

Intensive, project-based program covering data science lifecycle, from data wrangling to deploying machine learning models for business impact.

- Completed 20+ hands-on projects and 600+ hours of curriculum, mastering Python, SQL, ML, and cloud tools to solve real-world business problems.

BS Universidad Veracruzana, Architecture Mexico, Sept 2014 – Nov 2019

Bachelor's degree in Architecture.

Certifications

Data Science Methodology | IBM 2023

Mastered the CRISP-DM framework.

- Application of the end-to-end CRISP-DM methodology, used to structure ML projects from the definition of the business problem through to the deployment plan.

Python for Data Science, AI & Development | IBM 2024

Python foundation for data science and AI.

- Mastered Python programming (functions, OOP, error handling) and core data science libraries (Pandas, NumPy) for data manipulation and analysis.

Construction Project Management | Columbia University 2022

The fundamentals of construction management, LEAN projects and sustainability

- Mastered management for construction industry, the role of project manager, contract types, and project delivery methods.